

ActInf Textbook Group ~ Cohort 7 ~ Session 20 (Chapter 8) ~ 3/20/2025

TextbookGroup/ParrPezzuloFriston2022/Cohort_7 | Cohort 7 ~ Session 20 (Chapter 8) ~ 3/20/2025 | 56:52

Welcome everyone to the Active Inference

Institute textbook reading group. This is cohort 7 continuing on with chapter 8 of the textbook. We already know that we are in the second half of the textbook, meaning we're much more focused on practice uh rather than theory uh which we had in the first part. In chapter 2, we were introduced in chapter six, excuse me, in part two, we were introduced in chapter six to a recipe for building active inference models. So we're familiar with all of uh you know the core fundamentals including what is your system of interest or your agent or your generative model that you are looking at and then what is on the other hand uh to use Mark Solmes's words uh just as we identified a system of interest we have to figure out our not system that is what is not the system of interest what is the environment or other variables uh that you know exist in the world but are not um part and parcel with our agent. Right? So that's that means identifying what is our generative uh process or the environment or the not system. And so we are familiar with a lot of the core components. We look at things like hidden states, observations and uh actions or policies which are sequences of actions. And then we identify the variables of the models, the different parameters. And from there we just build

upward and we look at what kind of model is necessary for understanding our particular system of interest. And uh so we looked at chapter 7 which had to do with discrete state space models and that covered largely the hidden Markov model in the musician example and then built up to the POMDP or partially observable Markov decision-m process uh which is sort of the exemplar discrete state space model and that uh was illustrated through use of the T-maze example mouse choosing left or right. So we know already with the discrete state space that we if we followed up to the end of chapter 7, we can build models where there are discrete actions being taken like the mouse chooses the left or the right arm of the maze or it chooses to visit the informative cue along the way. But these ultimately amount to three different actions per time step. Um they are not a continuous state space. Right? So for those who are very familiar with anything from just general statistics linear regression up to machine learning and deep learning models that work with continuous state spaces often times we want to create models that can predict continuous values such as 37.99 as opposed to choose left in the teammates right and that also means being able to read continuous data such as from data streams IoT devices um financial data uh there's a plethora of different kinds of continuous uh data to look at. So

that's what brings us here to chapter 8 looking at continuous time models. These are arguably where active inference started. Um neuronal signals are usually measured continuously including from uh electronphilography or EEG uh devices as well as uh fMRI and looking at bolt signals as well as a variety of other physiological data including things like heart rate. So whenever we get into chapter eight, uh we turn to continuous state space models which the authors state are well suited for modeling physical fluctuations impinging on sensory receptors and for the continuous motion of the aectors, eg the muscles we used to change the world around us with model examples for the dynamical systems involved in motor control as well as the concept of generalized synchrony and finally constructing hybrid models for combining discrete and continuous variables at different levels which features at the end of the chapter. So largely we're introduced to notation for stochastic equations uh which invol are involved in continuous time models. Uh we see a lot of similar notation to what we've seen so far earlier on in the textbook. The authors largely use y to denote uh observations and x to denote hidden states. here. This is not terribly different, but we notice that we're also looking at and excuse me, let me go to this is technically the one for the generative model. Um, here we're also looking at a more continuous state

space. We're looking at something like a kind of error term or or or flux uh or or stochasticity within an equation as well as we're seeing a derivative. So we're looking at uh states evolving over time as a function of hidden states um as well as well or rather what is the rate of change of x given x and some kind of static or otherwise variable with stochasticity. So from there we're introduced to other notions. These are all very kind of standard ideas that of course are absolutely not unique to active inference. This is active inference building upon a lot of previous work. And so we're looking at an attractor system, right? So, so this uh generative model in equation 8.3, we're introduced to this idea that V is a kind of attractor. So whenever X is too high, it tends to become its rate of change becomes negative as it heads back towards the value of the attractor. and vice versa. If X is too low tends to increase back towards the value of the attractor. And so that means as an attractor is a kind of attracting or fixed point. Um, I I I emphasize this despite its simplicity because when you think about it, we're oftentimes trying to model systems that, you know, whenever it comes to agents or looking at organisms or anything that needs to persist in some kind of way within an environment, whether it means an organism trying to take care of its homeostasis or it be a thermostat reacting to changes in temperature, we

tend to be dealing with something like attractors. we we tend to deal with, you know, an attractor for me is that my body temperature tends to be roughly around 98 degrees Fahrenheit, as I use Fahrenheit in the United States. And uh so so we're looking at attractors in the sense of just where where do things tend around, right? Uh without the attractor, we can potentially completely disperse depending on how our model is set up. There's nothing to kind of guide us in our directionality. And whenever it comes to maintaining a bodily or physiological or other kind of system that requires maintenance at some homeostatic level or range, we can think of the homeostatic level or range as very much, you know, kind of an attractor or set of attractors uh in which the model tends towards to persist. Right? So, um we see a lot of these different kinds of dynamics play out. We're also introduced to precision, attention, and sensory attenuation. Many of these make sense to introduce here. Precision not and potentially attention. Uh they they could come in more deeply in earlier chapters and of course they do get mentioned in previous chapters, but essentially they make sense here because they tend to be viewed as continuous values. Not necessarily, but um so so they just relate much more to the material in this particular chapter with precision as a kind of sharpening of a probability distribution. So if precision is uh below zero or excuse me

between zero and one. So what can happen is that your probability distribution can come become more uniform and vice versa if your precision is far above or even a little bit above one then the precision of your probability distribution is going to sharpen right so a general sense of this is that if I have a strong amount of trust in the observations I'm seeing that is I can clearly tell that the thing that I'm seeing in the distance I feel fully confident it is what it is and we kind of liken that to the idea of precision that means there's a very high probability that the thing I see in the distance is what I think it is. The low near zero probability that it's going to be something else. Right? So we get this very sharp distribution where a lot of the the density is centered around one possibility. Right? Um similarly uh the other way around if precision is low we we see a much more flat broader distribution. I have no idea what that thing in the distance is and I'm not very clear on it. So precision in that sense is kind of like a sharpening or lessening dynamic. It can be related to synaptic gain uh in in neuronal dynamics. Um it it's it's seen as being very crucial for uh looking at things like confidence, right? Whether it be confidence in your your A matrix, your uh in terms of being able to uh interpret a situation or a B matrix in terms of like being able to look forward or back. And whenever we're looking at

continuous time models where we're no longer in the world of A and B matrices, although some of the dynamics are certainly similar in the sense of temporality that we find in a B matrix and relating uh observations or data or Y to hidden states or higher order beliefs or

X. So um we're also looking a lot at examples that use dynamical uh systems and so a good example is the generalized locker dynamics. This is kind of a classic ecological model that looks at uh relationships between plants, herbivores and car carnivores. The sort of common sensical notion of it is that the carnivores eat the herbivores who eat the plants and then the plants need enough time to regrow otherwise their population will completely die out. Um which of course because you can see the dependencies between these three different um organisms. They their their populations fluctuate over time as one proceeds to consume the other and yet they come back with reproduction and yet further if they were to to strongly diverge from these kinds of patterns.

What we we would have is eventual extinction of at least one of the species which necessarily in this situation would lead to extinction of the rest of the remaining species.

Right? So we see this kind of movement of them in kind of cyclical patterns nearly. And whenever we compare the herbivores and the plants or we compare the carnivores or the herbivores over

the course of playing out this simulation over time, we see that there is a kind of attracting shape to the continuous state space that is their populations or their joint um populations. or the way of looking at them as they correlate with one another in different ways over time. Um, we see these shapes play out as well whenever we look at the

Laurens system. And this is from the bird song example, which is where there are two birds who are singing with one another. And the idea is that they're going to kind of collaboratively finish a song where they take turns filling in space for the notes. I almost think of two guitarists playing solos or something and they're like trading off back and forth and yet they're still maintaining the same song. So there's a kind of synchronicity, the kind of rhythm that they need to maintain. Uh they need to stay on beat as it were. Um, and there's variation in what they play over time, right? They're not just both playing the same notes to where we get some very, you know, we don't settle in some clear equilibrium and stay there. There's a lot of variety going on and yet they have to stay in step. And um, ideally they are in fact playing the same song. So whenever we look at the Laurens system and we look at kind of what's going on in the state spaces here, we find attraction around two general areas of kind kind of like

two spheres of orbit, right? Um as these two birds sing their songs over time and then they continue to explore this state space as they go.

Right. So we see like or before excuse me let's go with this after learning they learn one another's patterns such that the state space they're exploring it tends to follow follow this line that is they're they're much more kind of synchronized and we notice relative to before learning where things were much more chaotic as they're still becoming attuned to one another's tunes. uh after learning not only is it more synchronized but we notice the axes have kind of increased right so in before learning they explore actually a significantly smaller area of the state space as their kind of correlated line between their actions you know spreads around after learning it's much more focused and they actually make it much further in different directions of the state space presumably they've learned to really well attune to one another while simultaneously being able to hit new notes that maybe weren't heard before. The notes presumably being part of the state space or identifiable therein. So a lot of this starts to look like when whenever we consider attractors but then we expand it to where there are multiple agents. It's interesting to see the different kinds of emergent phenomena that kind of arise out of the dynamics that play out between them and how they synchronize

because in active inference we've often talked about um agents who sort of that their internal states you know presumably if the the agent is is sort of optimal enough for the environment that they're in and find ways to adapt then what'll happen is that their internal states will somehow start to correlate strongly in in different ways and in different patterns. and perhaps with a temporality to where it's not as simple as drawing a singular um line but rather it'll be something much more complex like the dynamics for example of a Lorenz attractor whenever we we draw it out and how the state space looks from there. Um so all this is to say um much of this chapter while at first glance provides us with notation and the necessary equations for solving for minimizing free energy in various ways and continuous state spaces uh using those kinds of models. We're also along the way that the author authors cleverly snuck in all of these other aspects regarding synchronization, communication, some of the building blocks for thinking about multi- agent simulations. What does it mean for two systems, whether it be an agent and environment or it be two agents or otherwise? How do how do they synchronize one another? And then we're also reintroduced to the POMDP structure which we see is like everything in the upper half of this graph excuse me it is graph in figure 8.6 six um basing graph

and then the the upper half is the PMDP.
Everything below around where this dash
line is we can see is a lower level uh
faster time scale hybrid model in the
sense that for every iteration of the
POMDP up here we see that there are
actually three iterations of the
continuous state space model at the
lower level. So it's kind of like you
can imagine receiving a lot of data over
time and then at some threshold of time
steps such as three in this case we
could send that upward to the higher
level to say okay we receive this data
what does it mean right um it's kind of
like taking in the full
experience and then from there the
poomdp would then use its own model
parameters to update the different kinds
of inferences to send those back down as
priors to level one as we continue the
inference process. Um I
and we we have some nice discussion
again looking at
sakads and
um those kinds of dynamics which are
much more clearly continuous state
spaces more often than not when looking
at kind of like continuous coordinate
systems. Uh we also have mixture models
and clustering which again kind of uh
imagine for those unfamiliar with that
kind of logic imagine taking that hybrid
model we just saw and kind of smashing
it down such that we can receive um
continuous data but then use it to infer
clusters or discrete state space data.
Right? So you you see a certain kind of

pattern in the physiology and you determine that someone shows a particular discrete health state such as they are healthy versus another combination of continuous values their heart rate their glucose levels and so on. And in this particular state space uh discrete state space space they are they are um moderately unhealthy right and and with this set of values right so it's kind of going continuous to discrete

um so I think I'll leave it there because I've already talked for quite some time apologies for that if it took away from anyone um I'd like to open up the conversation then if anyone has any questions or anything in particular particular that stuck out to them in this chapter. And we can discuss it if you like.

There's always the um I'll make a first comment and then John or Isabella like in the first equation of the chapter it uh

But you said neural signals are measured continuously. So often times people look at the discrete spiking of a single neuron you know even though that's that's a simplification but measurements are often continuous and then later discretized. So it's sort of like if you look at the instrumentation that's actually coming off of real world sensors in a way they're kind of like analog and then digital systems discretize that for the for for

different kinds of advantages but um analog measurements are prevalent and then this is just the the um even though confusingly with the G and an F that don't map to equation 2.6 six and 2.5.

This is sort of a different look on the S and the O from the POMDP in terms of having uh observables Y which is O latent states X which is S and then instead of having the B transition matrix interpolated between different latent states rather there's a continuous um first derivative landscape with $\dot{X}$ so kind embodies the same uh or analogous elements but with different letters.

Yeah, thanks Daniel. Yeah, it's good points and because it's worth just really getting familiar more with the notation for anyone who's still kind of picking up on active inference and I with this in one way and I don't I don't want to these are my words not the author's words of course but in a sense if you think about the the A matrix as a in the in the discrete state space case in chapter 7 um and we think of how that is relating hidden states to observations. Then we do have something like a relationship here for this generative model that very much relates the two. And then if we think of a B matrix as the probability of the next state conditioned on the current state and uh the current action or policy

being pursued. Um this of course is different. We're looking at derivatives or rates of change. But nonetheless, we are essentially in this first example, we're relating hidden states and observations. In the second, we are relating hidden states to something like the temporal aspect of hidden states where the hidden state will come next. So of course, it's not a transitioning to a new state necessarily, but it's looking at its rate of change, which then of course could be used to predict what the value of the next hidden state will be in this continuous state space. And then we also have you know V almost like its own third variable and it itself doesn't have to be static. This of course is just like a nice simple example and it can be used as an attractor in this case. And then also with neuronal signals just with my own some of my own previous work on uh EEG signals it's so interesting in the sense of like a lot of them still are being uh recorded these signals are being recorded like digitally uh increasingly so and so you're kind of limited to the kind of rate at which the technology records and so even though you might looking at something down to the tenth of a millisecond. If we think about how when we take a derivative, we're technically like having to do a lot of um discretization of values in order to you know uh carry out differentiation or or integration. Then then similarly with

EEG it's like okay well I'm still discretized to the tenth of a of a nancond or or a of a millisecond or whatever you know rate it's recording at. So it is it is interesting like once you get down to these fine fine lines it's like what is discreet or continuous any anymore that's much more dramatic than than probably what the sentence that should actually come to mind but still it's uh there's more nuance but but still it's um it is interesting to think about and for all intents and purposes for the most part with with neurophysiological or any kind of physiological data you still are typically looking at continuous data unless unless there's some kind of particular criteria, right? That that that that someone makes a judgment about a value they see and then they just mark it as like, you know, on a liyker scale 1 to 5. It's still a discrete state space with an implied ordering to it. Uh for example, so anyway, but now we're talking a lot about a I'm talking a lot about data types now.

Um,
so see maybe something we could pick up on in the chapter 8 questions unless anyone
No. Okay, no problem. Yeah, let's look at chapter 8 of the KOD.

Nice summary that Daniel and Ally did a couple years ago.

Let's

see. Yeah. Another random thought is it's just interesting that the analytical relationships hold for the discrete case at all when they're more underlying differentiable flow-based form is continuous. And yet it works like almost exactly how one would think in the discretized setting kind of however you discretise it. So it's like underlying circle and then whatever resolution your screen is. It's like it kind of works out. But that doesn't seem guaranteed by the definition of a circle. But yet it is.

I know it's an interesting way to put it.

It's I mean

it it's funny. It sometimes it comes down to like making your own kind of decision whenever it comes to the modeling like do you want to view something as discrete? But then of course if you're going to build an active inference like model then it becomes insanely important to decide which state space you're working with right because we then we do start to see you know I mean underneath there's similar mechanics between the two kinds of models if we're able to just say oh there's just two kinds there continuous and discrete with all the infinite varieties in between I suppose but um but still we end up with with very different dynamics and also worth talking about then um I mean

I don't know no one's had any questions about it so I haven't gone into it but there are the generalized coordinates of motion which might just be but these are essentially sort of the the actual mathematical mechanics behind continuous state space models. Right? So it ends up what what ends up happening is that for those who are familiar with Taylor series approximation like you can view it as just you know you over you start to expand the number of of terms in your t uh Taylor series expansion. Um the the authors note that around six generalized coordinates are sufficient, meaning we're going to expand that far. Um and it works quite well not just as supplying these core functions and equations. And then furthermore, they're also very commonly used in analyzing neuronal signals. Um but many obviously many many many other types of data that don't have to have anything to do with neuronal anything necessarily but these are classic equations that you would find there and then it also works quite well in the sense of having a variational inference model right because as opposed to trying to like excessively predict into the future what will happen we can kind of simplify to or approximate a behavior often times as authors say with six generalized coordinates. Let's look at

yeah sensory attenuation. What are the causes and consequences of sensory attenuation? Just going to check the chat.

My understanding nothing more than stochasticity within the agent. Chaotic movement from one attractor or internal state to another. Movement and sensory attenuation. A point where you don't want to perfectly or accurately predict sensory consequences or associates of movement. Box 8.1. Belief I am not moving. Need to downweight the precision of that belief. Make it wider or lighter so that action shift can occur. policy shift occurs. Increase the attention again. Wow, we're somewhere between uh a sentence and a set of short causal sort of predicates here. Uh very symbolic, but actually we were getting warmer over time as we continued reading down these statements. This is not a bad way of compressing information that could help us to describe sensory attenuation.

Why is that sensory attenuation? So, one of the interesting things that can sometimes go overlooked in active inference and it's worthwhile to to know be aware of it for anyone who's interested in uh you know kind of understanding and then utilizing the active inference framework. We know that agents are self-evidencing, right?

So self-evidencing. Well, that means that in some way, if we just take it at face value, the idea is that an agent will

try to realize what it already expects,
right? We saw this in chapter 7 with
preferences. And the more expanded, you
know, way of looking at them is prior
over observations. And if they remain
fixed, which often times they do, such
as in the T-maze, that the mouse always
wants the cheese. At no point does it
start to want the shock and devalue the
cheese, right? So that reminds us of the
kind of um attractor dynamics, right?
You know, and so sometimes people have
used the phrase goal prior, which
probably should not be too terribly
confused with the phrase preference
prior, but it has been used in in
place of it to say that this is a prior
belief you have. And because it doesn't
change, that is where you are attracted
to, let's say. So whenever it comes to
sensory attenuation, the odd thing comes
up where in the moment you believe you
are sitting still and you decide you
would like to get up, right? But you
can't because you strongly believe
currently that you're sitting
still, but maybe you have in your goal
prior that you do or your preferences
you do want to stand up. But you can
imagine this kind of strange conflict
that occurs even though presumably all
of you wants in a more colloquial sense
to stand up. So what is this sensory
attenuation about and what is belief I
am not moving need to downweight the
precision of that belief so that action
shift can occur policy shift occurs
increase the attention

again let's go to

box

8.1

see of course it's a

text precision control is its role in

movement generation. To understand this,

it is worth thinking about what happens

in the absence of this control. Imagine

first that sensory data are predicted

with high precision. I am sitting down.

I notice I'm sitting down. The messages

from these data are therefore afforded

high synaptic gain and lead to various

inferences about the position of some

body part. I am definitely sitting down.

The problem with this is the equivalence

between motor commands and predictions

under active inference. An accurate

belief that I am not moving. I'm not

moving, I'm sitting down. Cannot be used

to predict the sensory consequences of

movement vital for the initiation of

that movement. How will I ever move if I

firmly believe I'm sitting

down? The idea is to attenuate or or or

sort of partially if not fully, I don't

want to say fully, but but to some

significant actionable degree to reduce

one's precision in their beliefs.

meaning that that that dynamic I

described earlier of sort of a a

flattening which is I they're implying

that in order to stand up I need to

actually temporarily attenuate my belief

that I'm sitting down just so that the

the proprioceptive signals and

everything else can play out such that I

can stand up right so so that there's

this kind of interesting philosophical
nearly aspect of this that is we have to
convince ourselves
implicitly. No assumption that any of
this is actually explicit in the sense
of fully conscious you're thinking this
out loud in sentences or otherwise. But
the sense is that you actually have to
kind of pause your beliefs in some way
or more literally it would be you have
to introduce uncertainty in your
understanding of the current situation
in order to actually change that
situation which is rather fascinating
and it it it comes up is so it's
something to potentially wrap your head
around. It's but it it starts to make
sense and then it becomes useful in a
variety of situations, right? For
example, um it's it's been found to be
rather significant in cases of
Parkinson's disease where someone is
struggling to attenuate
um their their sensory intake, right?
they're they're they struggle to to have
fuller control over how their body
responds, of course. Uh and and so it's
a kind of by being able to understand
sensory attenuation and its connection
to things like precision precision
control as well as being able to see in
other areas of the textbook how they
relate precision and in particular
precision values to potentially being
relatable to various
neurotransmitters. suddenly being able
to understand these kinds of dynamics at
least as of this point in time in the

textbook a couple years ago by these authors and the work has not stopped since so and I don't think it's going to but at least as far as computational psychiatry is concerned being able to make these kinds of connections that are interpretable and have neurobiological substrates involved um can at least hold the promise of being able to better serve different kinds of interventions and treatments for those who do struggle with these kinds of pathological disorders uh in ways that were not necessarily visible before because of a lack of the kind of underlying computational dynamics. And that's sort of why fields like computational psychiatry have been sort of taking off the past several years as the breadth of computational tools ever widens and more and more people produce open-source resources so that broader populations of people can get involved in these kinds of work. And of course they're not purely limited to computational psychiatry whatsoever. We could imagine anyone who's working in robotics who's working with a you know uh some kind of robot whether it be a automation system that's building car parts or it be like an actual robot that has limbs and navigates the world on its own. In either case, imagine if it's running on inference principles, especially a basian one where we can have the opportunity to have excessively strong priors such that the robot never does anything because it's too terribly

convinced of the current situation that it doesn't do anything to change it. Well, then suddenly sensory attenuation becomes one of the first concepts that you should look at. And it brings up interesting implications for neurotransmitters as well and relating them to other kinds of cognitive systems, right? Um being able to have the capacity for the synaptic gain precision control dynamics has a lot of implications for building any kind of predictive system. Whenever we're looking at this or similar frameworks, what is getting out of a chair? I was not the first. Um, yeah, it's also a sensory attenuation. It's like especially when we think about awareness or higher level introspection, we're attenuating almost all perception and only with really dedicated focus on sensory input can we even attend to you just all of our skin all of the sounds all these other things all spatial awareness ability to think of any concept it's like it's all being attenuated away that's sort of the attention question so it's like there's the attention is all you need and then there's the other side of the coin point is attenuation. But that's the same thing. It's just like whether you focus on upweing what is to be focused on or

downweing what
isn't. Yeah. Yeah, definitely. I mean,
it's like if I'm driving on the highway
in Chicago, as I occasionally do,
um, I mean, if I attempt to
maintain attention,
uh, whenever it comes to every single
car that's currently in my frame of
vision, I'm certainly going to have a
tougher time focusing on the particular
things I need to attend to, such as the
car immediately in front of me. that is
suddenly breaking and so I should
respond. And so it's just another way of
kind of reating Daniel's point that is
the kind of attentional dynamics needed
to optimally or at least suboptimally
but enough
to survive a situation let's say um like
those are there and we can't really do
without them. um so much of the sensory
data that we
have hypothetically available to us goes
very unnoticed, right? So, and even for
in the case of robotics like if you have
a robot who is you know putting together
sequentially different parts on a
manufacturing belt um it's not if it
takes in every single signal that it
receives it might not be able to hone
its attention in on the very particular
task at hand. Right? So,
um relates physiologically. We can think
of things like metabolism. We can think
about energy uh expenditure and and
saving um or
conservation. So if you took in every
single thing you ever saw in its full

entirety as if we had perfect, you know,
um um like picture book like memory um
photographic memory. There it is.
And I mean thinking about it
computationally, I mean that's an absurd
amount of data. It's going to be a lot
and if it all gets con, you know, um
stored in long-term memory. So it's just
it's it's such an it's interesting thing
about I mean I've always appreciated in
this chapter that despite the fact that
it's strongly centered on in name
continuous time models we actually sneak
in a lot of really interesting concepts
that are much harder to demonstrate or
describe until we reach this part of the
textbook. This is the
um this is where they
apply or look at luck of volterra
dynamics again but instead we're looking
at it in terms of this eyelink
conditioning problem related to the
neurobiological substrate underneath.
And we're looking at these kind of
classic ways of just looking at
conditioned stimuli, unconditional
stimuli. For those who've have
backgrounds in psychology, this will be
um almost annoyingly familiar.
Um, we can look at how expectations play
out, but then we also see these kind of
potentially
cyclic dynamics that play out that
necessarily have to be that way, right?
Or close enough to being that way such
that the system can persist. And then
over here, this was, you know, just as
they're simulating the system of

handwriting. This is kind of the way it plays out. And then it itself has a kind of near cyclical dynamic. Like we were to segment this, right? We see a nearly identical structures right next to each other in what from far away looks like a simple accidental scribble.

[Music]

This is a little bit different. to just do this.

But so I

notice still haven't had any particular questions and this

is a Julia

library called RX

Infern. And there's been a nice partnership between the developers of RX and fur and the institute for quite some time now. Um a rather active group has formed around it uh for anyone who is ever

interested and we're looking essentially at applying and learning and understanding as well as giving feedback to the developers for this package. Um, and I the reason why I bring it up here and now is that in my opinion, this offers some of the most interesting tools for looking at continuous time and such a cool website, right?

Um, continuous time models. And so for anyone who wanted to get into this chapter but are still becoming more familiar with active inference itself and are really just wanting to look at how to apply active inference in ways

that they're already familiar with maybe computationally. This website alone will show you a variety of ways to potentially accomplish what you want to accomplish. ranging from the various equations and how they draw out variational free energy as well as Beth approximation

um talking more about the underlying dynamics involved that we get a very good picture of them in the textbook but very significant on this website as well plus you get to kind of see it like in action being tailored for direct modeling. So they have an entire set of examples that it looks like they're still doing some modifications to their website but it's all here. So we have basic examples basian linear regression just being able to simply look at the relationship between fuel consumption and speed. Faster you go the more fuel you burn.

hidden Markov model. They've been expanding their set of use cases that arcs infer could be used for. And so now we're suddenly looking at a hidden markoff model as we saw in chapter 7. This is particularly a Roomba navigating different parts of a house. And it the hidden markoff aspect is it relates its its visual sensory receptor of what the floor looks like to a hidden state. That is its belief about what room is it it is in over time in our classic perception loop. This is only perception uh hidden markov

models. And then they're starting to increase even further lately.

support for POMDP models. um which is great and suddenly makes this library incredibly relevant because that would mean that it also covers the kinds of state space models that are covered in Briston's classic SPM

12 package for mat lab as well as PIMDP for Python which um the SPM does definitely allow for different kinds continuous state space dynamic analysis but RX infer out of the box is very much ready to go for creating all kinds of models. Here's the Gaussian mixture model I mentioned earlier where you can take in continuous values

Y and then use them

to

infer

discrete clusterings, right? that are close enough together such that we decide, oh, this makes up a distinct cluster.

Right? So, it's so it's just interesting to think about the the many many ways for for anyone who's familiar with deep learning. Some of this will just be second nature to already be thinking about these things. But whenever we think about active inference, we think about basian perspectives where we're always coming into a situation with a prior and thinking about different data types. And for anyone who's actually writing code and and producing models, what data they have available in the

potentially infinite ways that they could put things together, not just what data are you using and and and what is the breadth of the types, but also what are the inference processes between them and what kind of state space are they in? Are you going to discrete discretise something yourself because it makes sense to you, but maybe it won't to someone else who looks at the same model? They wouldn't approach it that way. Auto regressive models lagging a variable on itself.

I mean these are classic models whenever it comes to looking at forecasting and looking at like you know where where data is sparse and then suddenly you decide oh if I use the same data stream but just lag it on itself maybe it is a function of itself like it suddenly the the single data source you have becomes useful for predicting itself in that sense so then we can see that play out with observations and then the hidden state inference kind of drags or follows after that. So they have this kind of lag that occurs over time.

So of course with generalisation with continuous state spaces you know as opposed to feeling where like we're constricted to the ab c and ease of pomdp models here you could have I mean of course your matrices in the pomdps could be massive in the sense of you can

include a broad variety of of sensory
receptors or or or data streams coming
in. You could have a plethora of
different kinds of hidden states or
hidden state factors uh each with their
own discrete hidden states within. You
could make an incredibly complex model.
But but that said with with continuous
time models it often feels a little bit
more general in the sense of you could
have m you could have various parameters
some for the same thing some for
different things some which are highly
related to one another but computed a
bit differently and then at the end of
the day you're when you're learning them
it's just to somewhat follow the logic
of a deep neural network or otherwise
we're just looking at something
Like how do we compute weights? But how
do we do
it with respect to
minimizing variational free energy as a
functional of those parameters and the
data coming in? And furthermore, much
more distinctly, how do we look at the
rate of change in those systems? And
then here was the nice generalized
synchrony piece. How do
we synchronize with one another in our
daily activities? I say as someone with
a social sciences background. So,
um, great point with the neural network
analogy
and reinforcement
and the reward function as needed to
give a gradient
descendable strategy to optimize a

functional of
weights which don't directly correspond
to the semantics of the probability
distributions themselves. It's just hope
that there's enough of these tiny guys
firing that you can recapitulate at a
higher level the semantics of a target
distribution versus doing variational
free energy as a direct
gradient tractable landscape on the
probability distributions of interest
and how learning can have a
and then that learning rate is fixed or
you could learn the learning rate and
then how fast you learned that. It's
like the buck stops
somewhere in a given
implementation even though there are
like infinite dimensional
approximations.

It's what brings me back to
the just the general sense of an
attractor is well all these things could
move. We know from chapter six that a
lot of it as well as chapter nine I say
as we wrap up here but
uh chapter nine we also had that that
classic you know we're we're I'll just
say we're we're we're always making our
own decisions of how things should play
out and for one person learning a
learning rate is incredibly exciting and
a means of testing a very particular
hypothesis they have and then for
another person developing another kind
of system or not developing a system
rather just trying to create a
generative model that generally

accurately approximately approximates
uh an organism of interest that they
don't want to make any arbitrary
decisions. They want everything to be as
synced up to the organism they're
looking at as possible. The idea of
learning a learning rate becomes like
dangerous. Is it going to throw your
model results so far off or make them so
terribly
uh non-generalizable that it's like
why do any of this in the first place or
can you fit the model to find the
learning rate you know and that that
would be probably a more balanced
perspective. Um but in any case thanks
everyone
for joining today. uh really appreciate
it. And so a week from today, this same
time will be cohort 8, we'll be talking
about chapter three, I believe, the high
road of active inference. And then two
weeks from today at this time we'll move
on to chapter 9 as I was just mentioning
which we'll get much more into things
like model fitting and different kinds
of analysis and group
analysis and wrap up with models of
false inference. And that's very much
where the kind of computational
psychiatry line goes. But many others,
can a robot be
pathological? That kind of question
phrased exactly like that is ever more
strongly on the minds of many today. Um,
so interesting ways of thinking about
things, but um, yep, that's all for
today, folks. So, thank you very much.

Thank you, Andrew. Thanks, Isabella and John. Bye.